LinguaFranca: Causal Hop Localization Mid-Term Report

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Abstract

Agentic Retrieval-Augmented Generation (RAG) interleaves retrieval and reasoning for complex, multi-hop queries. As reasoning trajectories lengthen, models frequently fail—either by safely hedging or confidently hallucinating unsupported facts. While existing frameworks rely on post-hoc LLM evaluation to diagnose failures, we investigate whether these failures can be detected in real-time via the model's internal representations. We detail a validated data pipeline for generating controlled multi-hop failures in Llama-3.2-3B (Meta AI, 2024). We find that a linear probe trained on mean-pooled hidden states can detect hop-level failures with a peak overall accuracy of 92.5% (0.971 AUROC), detecting critical confident-wrong hallucinations with 90.2% accuracy. With the probing methodology validated against robust controls, we additionally finalize and validate a causal activation-patching pipeline, with preliminary pilot evidence that restoring clean hidden states at middle-to-late layers can reverse an induced hallucination; the full statistical evaluation across all counterfactual pairs is in progress and will be reported in the final submission.

1 Refined Problem Statement

The original proposal sought to detect a failing reasoning hop "at the boundary of the current hop, before subsequent reasoning steps are generated." Through empirical testing and pipeline development, we have sharpened exactly what this boundary detection entails. Our linear probe operates on hidden states mean-pooled over the generated text of a specific reasoning hop. Therefore, the refined detection target is evaluating: *"Does this stated entity match what is grounded in the context?"* rather than attempting to detect a pre-commitment signal of doubt before the hop is generated. While detecting pre-commitment signals remains a valuable avenue for future work, our current claim strictly bounds the probe's capability to evaluating the generated representation itself.

Furthermore, empirical analysis of our dataset revealed that "failures" are not a monolith. A significant portion of failure-labeled hops occur when the model correctly abstains from answering due to missing context (e.g., "there is no information about X"). We term this *hedging* (Table 3). This is a safe, desired behavior in a RAG system. The genuinely dangerous failure mode is the *confident-wrong* generation, where the model confidently hallucinates an unsupported fact. Our refined framing explicitly distinguishes between these two modes, recognizing that a viable failure-detection system must excel at catching confident-wrong hallucinations.

What Is Novel Here. Linear probing of hidden states and activation patching are both established techniques individually (Hewitt and Liang, 2019; Meng et al., 2022); our contribution is not a new probing or patching algorithm. The novelty is threefold and specific: (1) we are, to our knowledge, the first to apply internal-state probing at *individual reasoning-hop* granularity within a compositional chain, rather than at the query level (Section 2); (2) we identify and validate the hedging-vs-confident-wrong typology as a necessary axis for failure detection, which no cited prior probing or text-evaluation work makes; and (3) we are the first in this line of work to move beyond a purely correlational probe to a *causal* validation of the failure signal via activation patching, directly testing sufficiency rather than only predictive accuracy. Each piece individually is a known technique; their combination to produce a causally-validated, typologically-aware, hop-level failure detector is what we claim as the contribution.

Project Contributions & Mid-Term Status: Our proposed work outlined four primary contributions. Their mid-term completion status is as follows:

- **High-Fidelity Hop Labeling Engine (Completed):** We built a rigorous, automated 3-tier grading pipeline capable of assigning deterministic success/failure labels to unstructured reasoning hops at scale.

- **Counterfactual Hallucination Dataset (Completed):** We generated a highly controlled intervention dataset utilizing exact token-length entity swapping to force confident hallucinations without altering context.

- **Internal State Probing (Completed & Validated):** We trained linear probes across all Llama-3.2-3B layers, achieving 92.5% accuracy at Layer 10 and discovering the critical distinction between hedging and confident-wrong failures.

- **Causal Tracing (Pipeline Validated, Full Sweep In Progress):** We implemented a denoising activation-patching harness over our 115-pair counterfactual set, resolved the patching-direction and injection-position decisions described in Section 6, and validated the generation pipeline's correctness and determinism (deterministic greedy decoding, confirmed identical across repeated runs) on pilot examples. The full 115-pair $\times$ 28-layer sweep is currently running and will be reported in the final submission. Runtime repair integration (Phase 3, design detailed in Section 7) has not yet begun.

2 Literature Review

Our work sits at the intersection of Agentic RAG repair (Lewis et al., 2020), internal probing, and hallucination detection.

Trajectory-Level Repair: The Doctor-RAG framework (Jiao et al., 2026) provides our primary baseline comparison. It employs a "Diagnose-and-Repair" mechanism that waits for a full trajectory to fail before using a LLM to diagnose the text. Our work seeks to preempt this by shifting diagnosis to real-time internal state probing.

Query-Level Gating vs. Hop-Level Causal Diagnosis: Probing-RAG (Baek et al., 2025) is our closest prior work in substrate — it also probes intermediate hidden states of an LLM within a RAG pipeline. However, its prober answers a fundamentally different question: at the level of a whole query, after each retrieval round, is *another retrieval necessary*? Its probe is trained on a binary outcome label (whether the eventual generated answer matched the ground truth), iterated across retrieval rounds, and used purely as a retrieval-triggering gate. It makes no claim about *where* in a reasoning chain an error occurs, nor does it distinguish *why* a step failed. Our work instead probes at the level of an individual reasoning hop within a compositional chain, predicting not just whether that hop failed but whether the failure was a safe hedge or a dangerous confident hallucination ,a distinction Probing-RAG's query-level, outcome-only label cannot express. Furthermore, we validate this signal causally via activation patching (Section 6), demonstrating that the probed representation is not merely correlated with failure but is sufficient, when restored, to reverse it; Probing-RAG offers no such causal validation of its retrieval-gating probe.

Step-Level Text Evaluation: Process Reward Models (Lightman et al., 2024) represent the closest text-based method in terms of granularity, rewarding intermediate reasoning steps. However, PRMs require separate forward passes and expensive training, whereas we aim to extract this signal directly from the generative model's own hidden states.

Activation Patching: To establish causality, we adapt the activation patching and causal tracing methodologies of Meng et al. (2022) and Vig et al. (2020). This forms the basis of our Phase 2 objective to localize where the failure originates within the network.

Probing Methodology: We draw on rigorous probing controls established by Hewitt and Liang (2019), specifically utilizing shuffled-label control tasks to ensure our linear probes (such as our headline SVM) are extracting genuine reasoning properties rather than merely memorizing dataset artifacts.

The Gap. No work surveyed above combines all three properties our setting requires: (i) diagnosis at the granularity of an individual reasoning step rather than a whole query (unlike Probing-RAG), (ii) a failure-type distinction between safe hedging and dangerous hallucination (unlike PRMs or Doctor-RAG's trajectory-level diagnosis), and (iii) a causally validated internal signal rather than a purely correlational probe or a post-hoc external LLM judge (unlike Doctor-RAG and Probing-RAG alike). Our work is positioned to close this specific combination of gaps.

3 Methodology: Data Pipeline

To generate a heavily evaluated dataset, we built a custom pipeline over the 2WikiMultihopQA dataset (Ho et al., 2020). We filtered the raw dataset to isolate multi-hop compositional questions, oversampling to yield 4,115 valid evaluation examples containing 12,340 reasoning hops. The raw generative process yielded a natural failure rate of 43.93% (5,307 natural failures vs 6,774 successes).

To rule out the possibility that our probes might merely memorize entity-question pairs rather than internal uncertainty, we generated a perfectly controlled counterfactual dataset. For 115 success cases, we forced a "confident hallucination" by swapping the "entry entity" in the question itself (the entity that triggers the first reasoning hop) with a plausible distractor, leaving the RAG context unchanged (Table 2). These substitutes were rigorously constructed by drawing entities from other examples in the dataset, ensuring exact semantic type and token-length matching, with alignment verified against the model's tokenizer. Strict token-length matching is a mandatory prerequisite for Phase 2 activation patching, as it guarantees prefix-token alignment—ensuring that hidden-state patches land at the exact same semantic position across both the clean and corrupted prompts.

Including these counterfactual injections, the evaluation dataset totals 5,554 failure hops (Table 1). Of these failures, only 477 (8.6%) were safe "hedging" failures (e.g., "The context does not mention..."), while the vast majority (5,077) were dangerous "confident hallucinations." Several critical design decisions were made to ensure data validity:

Compositional-Only Filtering: We filtered the raw dataset to exclusively include compositional questions, discarding comparison-type questions. Unlike compositional questions where an error in Hop 1 causally propagates into the premise of Hop 2, comparison questions feature parallel, independent hops. Discarding them was a fundamental theoretical requirement to guarantee strict causal error flow for our Phase 2 activation patching, while additionally ensuring a consistent 2-hop structure.

Chain-of-Thought (CoT) Generation: We prompted the model to explicitly output a step-by-step reasoning chain (e.g., "Hop 1: [Fact]... Hop 2: [Fact]"). This discrete, stepped generation forced the model to break down its reasoning, allowing us to extract the hidden state "brain scans" precisely at the boundaries of individual hops.

Automated Success Labeling: To train our probes, we needed a ground-truth label (Success or Failure) for every single generated hop. Because models produce unstructured free-text, grading them automatically is notoriously difficult. We solved this with a two-part labeling engine:

1. *Bipartite Hop-to-Gold Alignment:* First, we aligned the generated text hops with the dataset's "gold" (correct) facts. Instead of assuming the model always generated the facts in perfect order, we used a greedy bipartite matching algorithm. This ensured that if the model resolved sub-facts out of order, we still graded it accurately. (Note: Our dataset contains 6,786 total success hops, slightly higher than the 6,774 natural successes. This discrepancy of 12 hops occurs because some Hop-2s within our 115 counterfactual examples still succeed naturally even after Hop-1 is forced to fail).

2. *3-Tier Entity Evaluation:* Once aligned, we determined if the generated hop actually contained the correct entity. We built a 3-tier system: (1) normalized string matching, (2) alias table resolution, and (3) SBERT semantic similarity (Reimers and Gurevych, 2019) (cosine threshold = 0.70). Only highly ambiguous candidates falling below the SBERT threshold were passed to a budgeted LLM judge(Llama-3.2-3B) as a last-resort check. This maintained cost control at scale while ensuring near-perfect label assignment.

Context Grounding: We explicitly restricted the prompt context to *only* the gold supporting passages (identified via the dataset's `supporting_facts` field), discarding all distractor passages present in the raw 2WikiMultihopQA records, and paired this with a strict system prompt instructing the model to rely only on the given context. This deliberate fix was implemented after discovering the model was defaulting to parametric memory and bypassing the RAG context entirely. If a model answers from its pre-trained memory, it can "cheat" and answer correctly even when the necessary facts are missing from the provided text, making it impossible to reliably induce and isolate RAG-specific reasoning failures.

Scope: Retrieval vs. Reasoning. We deliberately scope this mid-term submission to the *post-*

retrieval reasoning stage of a RAG pipeline rather than the retrieval stage itself. The retrieval component is simulated by directly supplying the gold passages (Context Grounding, above), rather than querying a live retriever over a Wikipedia passage pool. It isolates reasoning-time failures (the model misattributing or hallucinating an entity despite the correct evidence being present) from retrieval-quality confounds (the correct evidence not being found), which is a separate failure mode from the one we target in Phases 1–2. Live retrieval was always planned for Phase 3. Our design embeds the partially generated hop text with MiniLM (Reimers and Gurevych, 2019) to query a FAISS index (Johnson et al., 2021), fetching the top-k (k=3) Wikipedia passages *during* generation rather than upfront. We haven't implemented this yet simply because Phase 3 (retrieval-driven repair, Section 7) has not started, which strictly follows our original project schedule. Importantly, our current probing and patching results (Phases 1–2) do not depend on how retrieval is done, meaning they require no changes once live retrieval is added.

Table 1: Hop-Level Evaluation Dataset Statistics

Metric	**Count**
Total Compositional Questions	4,115
Total Generated Hops	12,340
Success Hops	6,786
Failure Hops (Total)	5,554
↪ Hedging Failures	477 (8.6%)
↪ Confident-Wrong Failures	5,077 (91.4%)

4 Methodology: Probing and Validation

To determine if the model internally "knows" it is hallucinating, we must analyze its brain activity during generation. For every token generated by Llama-3.2-3B (Meta AI, 2024) during a reasoning hop, we extract the high-dimensional vector representation (the hidden state) produced at each of the 28 transformer layers. Because a single hop consists of multiple tokens of varying lengths, we average these token-level vectors together (mean-pooling) to create a single, fixed-size representation for the entire hop. These mean-pooled vectors serve as the input features for our linear probes.

To ensure our probes were genuinely detecting failure and not just exploiting statistical loopholes, our validation sequence included several rigorous checks:

Initial Leak and Correction: Early un-grouped splits yielded a suspicious 100% probe accuracy at layer 27. We diagnosed this as a data leak (test examples sharing the same base question as training examples). We corrected this by implementing a Grouped Cross-Validation (GroupShuffleSplit) keyed on the original question ID, ensuring perfect separation. Specifically, for each of the 5 folds, our 12,340 evaluation hops were split strictly into 9,872 training hops and 2,468 test hops.

Metrics. Unless otherwise noted, *Overall Accuracy* and *AUROC* are computed on the binary Success/Failure label over all held-out hops in a fold; *Hedging Acc* and *Confident Acc* are accuracy restricted to the subset of held-out hops whose true label falls in that failure subtype. All reported figures are the mean across the 5 grouped CV folds unless a single-fold or single-layer figure is explicitly stated; per-fold standard deviation is not yet reported in this draft and will be added alongside the full causal patching sweep in the final submission.

Grouped Cross-Validation Results: Under strict 5-fold grouped CV, we compared Logistic Regression, Linear SVM, and MLP classifiers (Pedregosa et al., 2011) across all 28 layers of the model. We found that performance was highly consistent across probe architectures, indicating the signal is deeply linearly separable. The Linear SVM proved optimal: it marginally outperformed Logistic Regression (92.5% vs 92.4%) while running more than twice as fast during inference ($\sim$0.02ms vs $\sim$0.05ms per hop). The MLP(92%) (1 hidden layer, 128 dimensions) offered no statistically significant accuracy gains over the linear models (scoring within 1 standard deviation across all 5 folds) despite higher computational overhead.

Across the 28 layers, the probe accuracy demonstrated a distinct rise-plateau-decline shape (Figure 1). The plateau occurred across layers 9-13, with the Linear SVM at Layer 10 peaking at an exceptional **92.5% overall accuracy** and **0.971 AUROC** (Table 4).

Shuffled-Label Control: To rule out the possibility that the probe was simply fitting noise on a small dataset, we ran a shuffled-label control (Hewitt and Liang, 2019) specifically on our headline SVM model. The probe's accuracy immediately collapsed to 0.506 (random chance), proving the extracted signal is genuine.

Table 2: Qualitative Example: Induced Confident-Wrong Hallucination via Counterfactual Swap

Component	Text
Context	*...Inception is a 2010 science fiction action film written and directed by Christopher Nolan...*
Original Q	Who directed Inception?
CF Swap	*Inception* $\rightarrow$ *Avatar* (Matched tokenizer length)
Poisoned Q	Who directed Avatar?
Generation	**Hop 1:** Avatar was directed by Christopher Nolan.
Failure Type	Confident-Wrong (Entity mismatch flagged by 3-Tier Grader)

Table 3: Qualitative Example: Natural Hedging Failure

Component	Text
Context	*...The band released their debut album in 2011, but chart positions were not recorded...*
Question	What was the peak chart position of the debut album?
Generation	**Hop 1:** The context does not mention the peak chart position.
Failure Type	Hedging Failure (Safely admitted missing information)

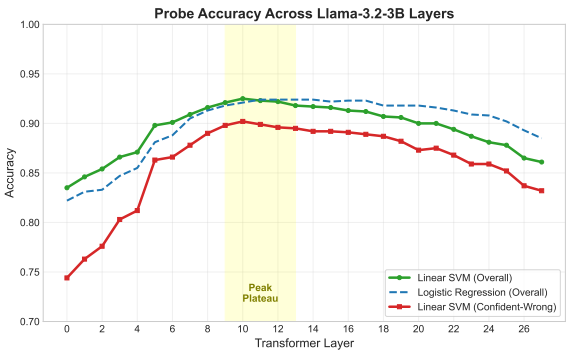


Figure 1: Probe accuracy across Llama-3.2-3B layers. The curves exhibit a clear rise-plateau-decline, with the confident-wrong signal heavily trailing the overall accuracy but peaking simultaneously in the mid-layers (9-13).

Hop-1-Only Control: To ensure the probe wasn't trivially exploiting hop positional encodings (i.e., learning that hop 2 fails more often than hop 1), we trained and tested the SVM exclusively on Hop 1 data. The probe achieved 0.951 accuracy and 0.953 AUROC against a 0.916 majority baseline. AUROC is the primary headline metric here due to the heavy class imbalance in Hop 1.

Hedging vs. Confident-Wrong Split: Our most critical validation involved splitting the failure class. Hedging failures exhibit near-trivial lexical patterns. The harder, more meaningful target is the confident-wrong subset. We found that while hedging accuracy remains near 1.00 across all layers (peaking at 96.7% for the SVM), confident-wrong accuracy is meaningfully lower but still exceptionally strong in the mid-layers (Layer 10 SVM: **90.2% accuracy on the confident-wrong subset**).

Baselines: We evaluated three non-neural baselines. The TF-IDF text baseline achieved 0.716 acc / 0.796 AUROC. Interestingly, a Hop-Idx-Only baseline achieved a surprisingly strong 0.783 acc / 0.855 AUROC. This is explained by the fundamental difficulty difference between hops: Hop 1 is typically an explicit entity ID extraction, whereas Hop 2 requires a harder attribute lookup, making position alone a strong predictive feature.

5 Baseline Comparison: LLM-as-Judge

To establish our RQ3(evaluating the comparative efficacy of internal probes versus text-based diagnosis in accuracy, latency, and cost) baseline against Doctor-RAG-style frameworks, we evaluated whether un-finetuned LLMs could diagnose failures post-hoc. The judge was provided the full context, the question, and the generated hop. We employed a direct next-token logit-scoring mechanism (extracting the raw internal probabilities of the 'Yes' vs. 'No' tokens) paired with a highly explicit system prompt.

As our primary baseline, we selected Llama-3.1-8B-Instruct (Meta AI, 2024). This choice serves

Table 4: Linear SVM Probe Accuracy Across Selected Layers

Layer	Overall Acc	AUROC	Hedging Acc	Confident Acc	Latency
Layer 0 (Early)	83.5%	0.914	95.3%	74.4%	0.024ms
Layer 5 (Early-Mid)	89.8%	0.953	96.2%	86.3%	0.023ms
Layer 10 (Peak)	**92.5%**	**0.971**	**96.7%**	**90.2%**	**0.023ms**
Layer 15 (Late-Mid)	91.6%	0.967	96.0%	89.2%	0.022ms
Layer 20 (Late)	90.0%	0.956	95.0%	87.3%	0.022ms
Layer 27 (End)	86.1%	0.928	91.6%	83.2%	0.022ms

two crucial experimental purposes. First, it yields the highest, most balanced accuracy among the calibrated options (87.0% overall), presenting the strongest possible opponent for our internal probe to beat. Demonstrating that the probe remains vastly superior (92.5% vs 87.0%, and <0.1ms vs 679.8ms) against a highly capable model is a far more convincing defense than beating a weak or miscalibrated baseline. Second, it is genuinely decoupled from the generator: the reasoning hops were generated by Llama-3.2-3B-Instruct, while the 8B model is a different checkpoint from a different training run. This decoupling directly addresses concerns of self-evaluation bias and aligns perfectly with Doctor-RAG's design principle of using a larger, separate model as a diagnoser.

Comparing the decoupled Llama-3.1-8B judge (87.0% accuracy) against a Llama-3.2-3B self-judge (81.4% accuracy) reveals a meaningful improvement from both scaling and decoupling, mirroring the exact effect that Doctor-RAG's architecture argues for. As a secondary baseline, we evaluated Qwen2.5-7B-Instruct (Team, 2024). While it did not surpass the 8B model in accuracy (78.9%), it proved exceptionally fast (163.2ms) and demonstrated that text-based detection capabilities hold across entirely different model families.[1]

While these baseline results demonstrate that text-based detection *can* work with a sufficiently calibrated and decoupled prompt, it remains vastly inferior to internal probing in efficiency and peak performance (Table 5). Our best Linear SVM probe achieves a robust accuracy advantage (92.5% vs 87.0%) over the strongest LLM baseline. More critically, it offers a massive speedup (<0.1ms vs 679.8ms per hop) and operates at zero additional token cost.

[1]We excluded Qwen2.5-3B-Instruct from primary consideration; the smaller judge collapsed into a degenerate always-reject policy (an extreme skeptical bias), rendering its artificially high sub-class recall rates untrustworthy.

6 Causal Validation via Activation Patching

While the probing results in Section 4 establish that hop-failure information is linearly decodable from hidden states, decodability alone is correlational: a probe could in principle exploit a feature that correlates with failure without that feature being causally responsible for it. To test causality directly, we adapt the activation-patching ("denoising") paradigm of Meng et al. (2022) and Vig et al. (2020).

Protocol. For each of the 115 counterfactual minimal pairs, we (1) run the clean prompt and cache the hidden state at the final token of the generated `<hop1>` span, at every layer; (2) run the counterfactual prompt unmodified to obtain a baseline (hallucinated) generation; (3) for each layer l, re-run the counterfactual prompt while overwriting only layer l's hidden state at the same token position with the cached clean state, leaving every other layer corrupted; and (4) check whether this single-layer injection restores the correct (clean) answer. We resolved the two open methodological questions as follows:

- **Patching Direction:** We adopt *denoising* (patching clean states into a corrupted run) rather than noising, since our research question is whether the clean hop-1 representation is *sufficient* to rescue a forced hallucination, directly mirroring the deployment scenario we target (intervening on a failing generation).
- **Injection Position:** We intervene at the final token of the `<hop1>` span (the entity-mention boundary), consistent with the single-token extraction used by our probes, rather than the last prompt token.

Restoration is scored via token-overlap (subset) matching between the generated and expected answers, rather than strict substring containment, to

Table 5: Comparison of Post-hoc Text Evaluation vs. Internal Probing. Calibration indicates whether the LLM exhibited a balanced decision boundary. The probe operates directly on internal states, requiring zero generated tokens.

Model	Overall Acc	F1	Hedging	Confident	Latency	Tokens	Calibration
Llama-3.1-8B	0.870	0.857	0.883	0.898	679.8ms	1096	Balanced
Llama-3.2-3B	0.814	0.795	1.000	0.821	275.7ms	1096	Balanced
Qwen2.5-7B	0.789	0.800	0.974	0.969	163.2ms	1129	Balanced
Qwen2.5-3B	0.569	0.667	1.000*	0.992*	352.6ms	1129	Biased ("no")
Probe (best layer)	**0.925**	**–**	**0.967**	**0.902**	**<0.1ms**	**0**	**–**

* Qwen2.5-3B's sub-class accuracies are artificially inflated by its near-universal failure prediction, rather than genuine detection capability.

credit valid paraphrases (e.g. "Munich, Germany" vs. "Munich, Bavaria, Germany"). We verified the pipeline's determinism by re-running identical code on the same inputs and confirming identical per-layer restoration outcomes, ruling out GPU-level generation noise as a confound before computing the full sweep.

Status. The patching pipeline is implemented and validated: the methodological decisions above are resolved, the generation pipeline produces coherent (non-blank) outputs under both the baseline and injected conditions, and determinism is confirmed (see above). On pilot examples, denoising restoration occurs at a scattered span of middle-to-late layers and not in the earliest layers, which is directionally consistent with the probing results in Section 4 (classification accuracy peaks at layers 9–13), though we do not yet treat this as a confirmed finding. The full aggregation across all 115 pairs and 28 layers is computationally intensive (115 pairs $\times$ 28 layers $\times$ 2 generations per layer) and is currently running; the resulting per-layer restoration-rate table and figure, and the comparison against the probe's peak layers, will be reported in the final submission rather than this mid-term draft.

7 Planned Repair Mechanism (Phase 3)

A natural objection to any restart-based deployment is that a plain retry regenerating the same hop from the same context with the same model — will simply reproduce the same failure, since nothing about the generation has changed. Our design avoids this by changing the *evidence*, not just re-asking the same question: on a probe-flagged failure, we retrieve fresh evidence before regenerating, so the regeneration call is not identical to the original one.

As originally proposed, Naive Restart integrates the trained probe into a live, custom token-by-token generation loop (implemented via direct calls to `model.forward()`, since generation must be paused mid-sequence). At each hop boundary, the hop's hidden states are mean-pooled and passed through the trained probe to obtain $P(\text{failure})$; if this exceeds a threshold τ calibrated on the validation split to maximize $F_{0.5}$ (penalizing false-positive interruptions, since halting generation unnecessarily is costly), generation is interrupted. Retrieval is then driven by the hop's *textual* content, not its hidden state (the hidden state is used only for the failure decision): the partially generated hop text is embedded with all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) and used to query a FAISS index (Johnson et al., 2021) over a Wikipedia passage subset, retrieving the top-k (k=3) chunks. Rather than splicing this evidence into the existing KV cache mid-generation , which risks silent attention-coherence degradation we flush the cache entirely, prepend the retrieved evidence to the original prompt, and restart the full forward pass. This is what "Naive" refers to: the restart is architecturally simple (full recomputation, no cache surgery) while still being different from a blind retry, since it is conditioned on newly retrieved evidence rather than repeating the exact same generation call.

We note this is deliberately simpler than Doctor-RAG's own repair taxonomy (Jiao et al., 2026), which distinguishes reasoning-error repair (regenerate only the flawed step, keeping prior evidence) from retriever-error repair (reformulate the query and re-retrieve). Doctor-RAG is used in this project strictly as an independent comparison baseline (Section 5, RQ3), not as a component our own repair mechanism is built from; a single, uniform retrieve-and-restart policy is our own proposed design and is evaluated on its own terms against that

baseline. This mechanism requires both the live MiniLM+FAISS retrieval step and the custom pausable generation loop described above.

8 Limitations

While our internal probe demonstrates exceptional accuracy, several limitations constrain our current findings:

- **Counterfactual Volume:** Our highly controlled counterfactual dataset contains 115 examples. While sufficient for the Phase 2 causal patching sweep, this modest volume limits the statistical precision of the per-layer restoration rates reported in Section 6, and wider confidence intervals should be assumed at the per-layer level.

- **Scope of Detection:** Our probe successfully detects whether the model's generated entity contradicts its context (a mismatch). However, it does not detect "pre-commitment doubt" before generation begins. Consequently, our planned repair mechanism (Phase 3, Section 7) triggers *after* a hop is generated, rather than preempting it.

- **Binary Detection vs. Typological Classification:** Our probe is currently trained as a binary classifier (Success vs. Failure). While it can successfully detect *that* a failure occurred, it does not currently classify the *root cause* of the failure (e.g., distinguishing whether the error stems from flawed retrieval, a logical reasoning breakdown, or a lack of evidence).

- **Generalization:** The current results are highly specific to the 3B parameter class and the 2WikiMultihopQA dataset layout. Extending these claims to larger frontier models or demonstrating out-of-domain robustness on HotpotQA (Yang et al., 2018) remains explicitly reserved for future work in Week 3.

- **Restoration Matching Criterion:** Our patching evaluation scores an answer as "restored" if every word of the expected answer appears in the generated output. This permissive, subset-based criterion was chosen to tolerate valid paraphrases, but could in principle overcredit a generation that mentions the expected entity in a negated or hedged context. Given the short, atomic nature of our target answers, we consider this a low-risk approximation, but note it as a precision caveat rather than treating the restoration numbers as exact string matches.

9 Timeline for Remaining Work

Our timeline has shifted slightly to accommodate the rigorous debugging and validation rounds, which were necessary to ensure the integrity of both the probe and the causal patching pipeline. The causal patching pipeline (Phase 2) is methodologically finalized and validated; the full statistical sweep is in progress. The remaining tasks are mapped as follows:

- **Week 1 (Early Oct) — In Progress:** Patching direction, injection position, and matching criterion are resolved and determinism is independently verified (Phase 2 methodology); the full causal tracing (denoising) evaluation across all 115 counterfactual pairs and 28 layers is currently running and will be finalized this week.

- **Week 2 (Mid Oct):** Implement Phase 3 as originally scoped: the custom pausable generation loop with probe-threshold interruption, the MiniLM-embedding + FAISS live retrieval step, and the full-restart repair mechanism (all detailed in Section 7); evaluate the practical efficiency gains of early failure detection.

- **Week 3 (Late Oct):** Conduct the final comprehensive evaluation on the 2WikiMultihopQA test split and measure out-of-domain generalization on HotpotQA.

- **Week 4 (Oct 25 - 31):** Compile final findings and format the final project submission (Due Oct 31).

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